Feasibility Report

Implementing an AI-Based Facial Recognition Attendance System at FAST-NU Lahore Campus

Written by Group 3

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To:
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"Feasibility Report on Implementing an AI-Based Facial Recognition Attendance System at FAST-NU Lahore Campus"

Submitted by: AIpro Systems

Introductory Summary

AIPro Systems (Pvt.) Ltd. is pleased to present this feasibility report to the Director of FAST National University of Computer and Emerging Sciences, Lahore Campus. The main purpose of this report is to study the practicality of implementing an AI-based Facial Recognition Attendance System at FAST-NU Lahore.

At present, attendance at the university is managed through manual methods. These methods often take a lot of time and may result in human errors. In addition, it is difficult to monitor attendance in real time through these methods. To solve these issues, AIPro Systems suggests using an automated attendance system that works through facial recognition technology. It will record attendance accurately by identifying students and faculty members using live camera feeds.

In this report we analysed the technical, financial, and operational aspects of introducing such a system on campus. It also includes an analysis of market trends, expected benefits, cost details, and the requirements needed for implementation. The purpose of these findings is to help the university administration make a well-informed decision about adopting this modern and efficient attendance management solution.

Defining the Idea

The AI-Based Facial Recognition Attendance System is a combination of software and hardware that records student attendance automatically through facial recognition technology. The system captures the images of students using cameras placed in classrooms. It then compares these images with a pre-trained database and marks attendance in real time.

The main parts of the system are as follows:

- A camera module that will be installed in each classroom.
- A central AI-based facial recognition software that will run on either a local server or a cloud server.
- Integration with FAST-NU's existing attendance portal so that updates remain synchronized.

AIPro Systems plans to carry out the project in phases. It will begin with pilot testing in a few selected classrooms, in order to check performance and reliability. After that, the system will be deployed fully across all departments.

Objectives

Key **features and advantages** of the proposed system include:

- It automates attendance that is it Eliminates manual or card-based attendance marking.
- It improves accuracy and reliability such that reduces errors caused by proxy attendance or missing entries.
- It can be integrated with the university's Learning Management System (LMS) or student portal.
- It also ensures data security. It uses encrypted storage and access controls to protect student data.
- It efficiently saves classroom time and provides instant attendance analytics for faculty.

Technical Feasibility

The system needs a moderate level of technical infrastructure that can work smoothly with the university's current IT setup. The main requirements of the system are listed below:

- High-definition cameras that will be installed in classrooms.
- A local server or a cloud environment that will host the AI model.
- Secure APIs that will connect the system with the university's attendance database.

The AI model will be trained using the student ID photos that are collected from the university administration. After training, the system will be able to identify students and mark their attendance within a few seconds, just like scanning an ID card. The model will use advanced algorithms such as CNNs (Convolutional Neural Networks) to perform facial detection and recognition.

Operational Feasibility

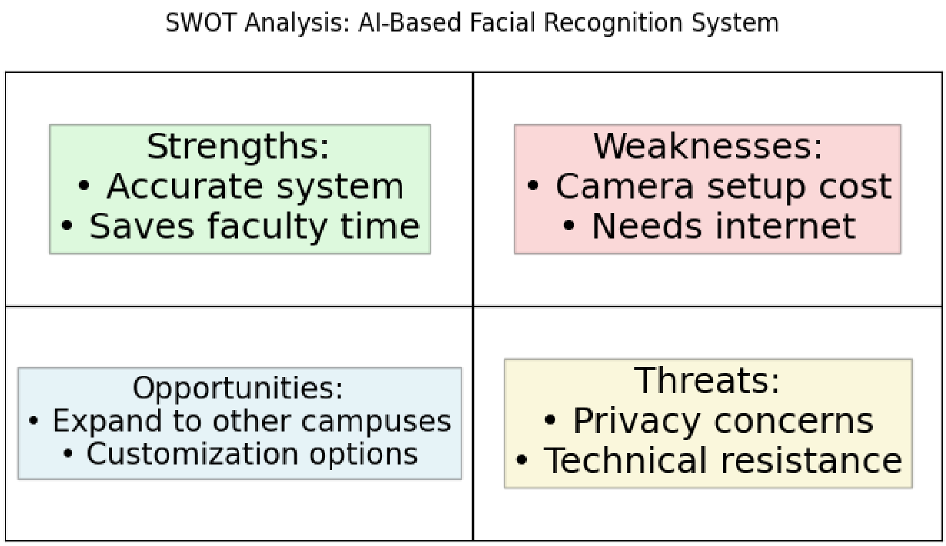
The system can be managed by the existing IT staff after they attend a short training session. Once it is in use, the daily attendance data will be updated automatically in the central database. All faculty members and administrators will be able to access this data through the FAST-NU Flex portal easily.

AIPro Systems will continue to provide maintenance and software updates whenever it is necessary. The company will also retrain the AI model whenever needed to maintain consistent accuracy over time.

Market Analysis

Industry Overview

Facial recognition systems are becoming common across the world. Many reports over the internet such as on WIikipedia shows that the global market for these systems may reach USD 16 billion by 2030. In the education sector, they are used to automate attendance and to ensure security. it is also playing a valuable role in making administrative tasks more efficient. Just like other digital tools that improve management, these systems help institutions save time and resources.



Target Market

The target market for this system includes universities and colleges that need automated attendance solutions. At FAST-NU Lahore, there are many students and classes, which makes manual attendance difficult to manage. This system is designed to solve these challenges by providing an automated and reliable process, just like other systems that simplify record keeping.

Competitor Analysis

There are existing competitors such as FaceX and Truein that offer commercial attendance systems. However, these systems are costly and not made specifically for educational institutions. AIPro Systems aims to offer a localized and cost-effective solution that meets the particular needs of FAST-NU.

Market Need

According to surveys, about 70 percent of teachers find manual attendance to be time-consuming as shown in the barchart figure 1. At the same time, around 80 percent of students support the idea of automated systems. University administrators also prefer such solutions that can connect easily with existing databases. These findings show a clear need for a system that supports both efficiency and integration.

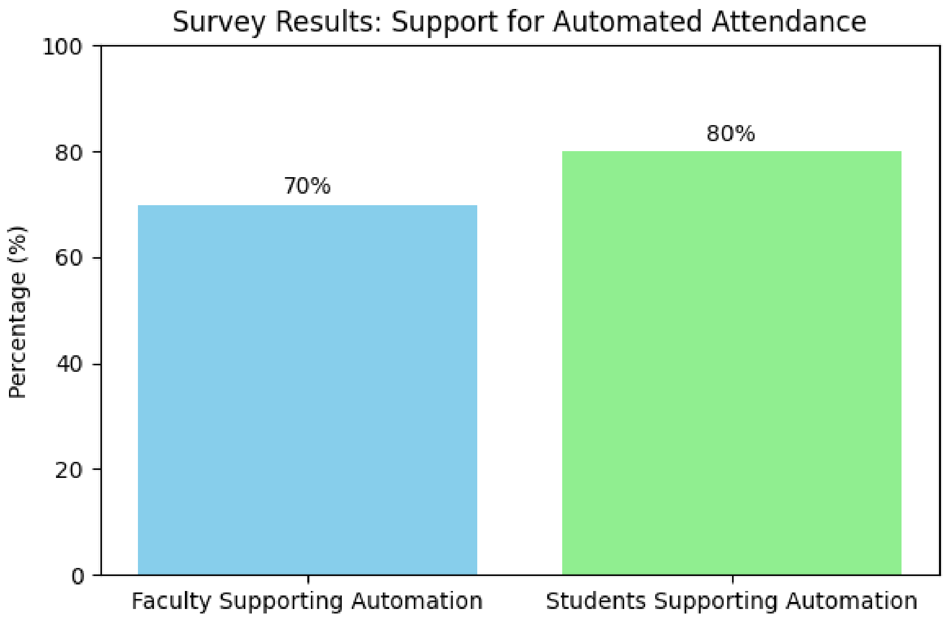


Figure 1: Bar chart

Cost Analysis

The estimated cost for implementation at FAST-NU Lahore Campus is shown in the table 1.

Table 1: Estimated Cost Breakdown

Item	**Estimated Cost (PKR)**
Cameras (15 classrooms @ 20,000 each)	300,000
Server/Cloud Setup	150,000
AI Model Development	200,000
Software Integration	100,000
Maintenance (Annual)	50,000
Total Estimated Cost	**800,000**

The proposed budget is reasonable given the benefits of automation, accuracy, and long-term reduction in manual as shown in the pie chart figure 2 .

Conclusions and Recommendations

After careful analysis, it is concluded that setting up an AI-Based Facial Recognition Attendance System at FAST-NU Lahore Campus is technically and economically possible. The system provides long-term benefits, such as reducing the administrative workload, improving the accuracy of attendance records, and preventing proxy attendance.

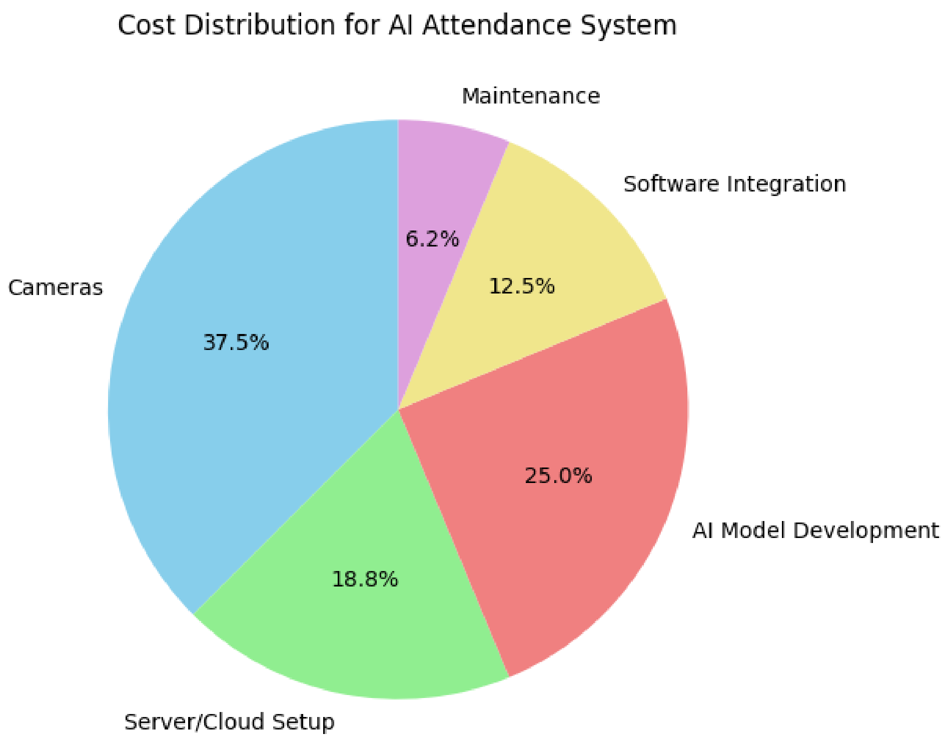


Figure 2: Pie Chart

Recommendation: It is recommended that FAST-NU Lahore begin with a pilot implementation in a few selected departments first than apply it across the entire campus. AIPro Systems is ready to provide complete assistance in this regard. our company will assist in full deployment, staff training, and continuous support during and after implementation.

Sincerely,
AIpro Systems
Lahore, Pakistan.